Warm-Up Quiz — Day 9

Chain Rule, the Normal CDF, and Score Functions

Complete before lecture. 10 minutes.

Name: ______________________________

Today we study probit, which models $P(Y = 1|X) = \Phi(X'\beta)$. This requires the chain rule for compositions with the normal CDF Φ and its derivative ϕ, and computing gradients of log-likelihoods (score functions).

1. **The Standard Normal PDF and CDF.**
Recall $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ and $\Phi(z) = \int_{-\infty}^{z} \phi(t)\, dt$, so $\Phi'(z) = \phi(z)$.

   (a) Compute $\phi(0)$.

   (b) Using the chain rule, compute $\dfrac{d}{d\beta}\, \Phi(3\beta)$.

   (c) More generally, if $\boldsymbol{x}$ is a fixed vector, compute $\dfrac{\partial}{\partial \beta_j}\, \Phi(\boldsymbol{x}'\boldsymbol{\beta})$.

2. **Derivatives of $\log$ Compositions.**
Recall: $\dfrac{d}{du} \log u = \dfrac{1}{u}$, so by the chain rule $\dfrac{d}{dx} \log g(x) = \dfrac{g'(x)}{g(x)}$.

   (a) Compute $\dfrac{d}{d\beta}\, \log \Phi(\beta)$.

   (b) Compute $\dfrac{d}{d\beta}\, \log\bigl(1 - \Phi(\beta)\bigr)$. (Hint: the chain rule gives a minus sign.)

3. **Probit Log-Likelihood.**
For a single observation $(Y_i, \boldsymbol{x}_i)$ with $Y_i \in \{0, 1\}$, the probit log-likelihood contribution is:

$$\ell_i(\boldsymbol{\beta}) = Y_i \log \Phi(\boldsymbol{x}_i'\boldsymbol{\beta}) + (1 - Y_i) \log\bigl(1 - \Phi(\boldsymbol{x}_i'\boldsymbol{\beta})\bigr)$$

(a) What does ℓ_i simplify to when $Y_i = 1$? When $Y_i = 0$?

(b) Using your answers from Questions 1 and 2, compute the score $\frac{\partial \ell_i}{\partial \boldsymbol{\beta}}$ for the case $Y_i = 1$.

4. Marginal Effects (Chain Rule).
In probit, $P(Y = 1|\boldsymbol{x}) = \Phi(\beta_0 + \beta_1 x_1 + \beta_2 x_2)$. The marginal effect of x_1 is:

$$\frac{\partial\, P(Y = 1|\boldsymbol{x})}{\partial x_1} = \quad ?$$

(a) Compute this derivative.

(b) If $\hat{\beta}_1 = 0.5$ and $\boldsymbol{x}'\hat{\boldsymbol{\beta}} = 0$ (so $\phi(\boldsymbol{x}'\hat{\boldsymbol{\beta}}) = \phi(0)$), what is the marginal effect?

Answer Key — Day 9

1. (a) $\phi(0) = \frac{1}{\sqrt{2\pi}} \approx 0.399$.

(b) $\frac{d}{d\beta}\Phi(3\beta) = \phi(3\beta) \cdot 3 = 3\phi(3\beta)$.

(c) $\frac{\partial}{\partial \beta_j}\Phi(\boldsymbol{x}'\boldsymbol{\beta}) = \phi(\boldsymbol{x}'\boldsymbol{\beta}) \cdot x_j$.

2. (a) $\frac{d}{d\beta}\log\Phi(\beta) = \frac{\phi(\beta)}{\Phi(\beta)}$.

(b) $\frac{d}{d\beta}\log(1-\Phi(\beta)) = \frac{-\phi(\beta)}{1-\Phi(\beta)}$.

3. (a) $Y_i = 1$: $\ell_i = \log\Phi(\boldsymbol{x}_i'\boldsymbol{\beta})$. $Y_i = 0$: $\ell_i = \log(1-\Phi(\boldsymbol{x}_i'\boldsymbol{\beta}))$.

(b) For $Y_i = 1$: $\frac{\partial \ell_i}{\partial \boldsymbol{\beta}} = \frac{\phi(\boldsymbol{x}_i'\boldsymbol{\beta})}{\Phi(\boldsymbol{x}_i'\boldsymbol{\beta})}\boldsymbol{x}_i$.

4. (a) $\frac{\partial P}{\partial x_1} = \phi(\beta_0 + \beta_1 x_1 + \beta_2 x_2) \cdot \beta_1$.

(b) $\phi(0) \cdot 0.5 = \frac{0.5}{\sqrt{2\pi}} \approx 0.199$.